Electrostatic Force

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Electric Charges

- There are two kinds of charges in nature, which were given the names **positive** and **negative**.
  - *The charge like the one of* ***electrons*** *is called* ***negative*** *and*
  - *The one like the charge of* ***protons*** *is called* ***positive****.*
- Charges of the **same sign repel** each other and charges of **opposite signs attract** each other.

Electric Charges

- Electric **charge** is **conserved**. That is, charge is not created or destroyed.
- Electric **charge** is **quantized**.
  - *That is, it can only assume **quantum values**.*
  - *These are multiples of a fundamental charge* $e = 1.60219 \times 10^{-19}$ *Coulomb.*
  - *This is the **charge** of the **proton** which is also the **absolute value** of the charge of the **electron**.*

Point Charge

- The term **point charge** refers to a particle of zero size that carries an electric charge.
  - The electrical behavior of electrons and protons is well described by modeling them as point charges.
  - Practically speaking any charged particle whose dimensions are negligible as compared to its distances from other particles can be considered to be a point charge.

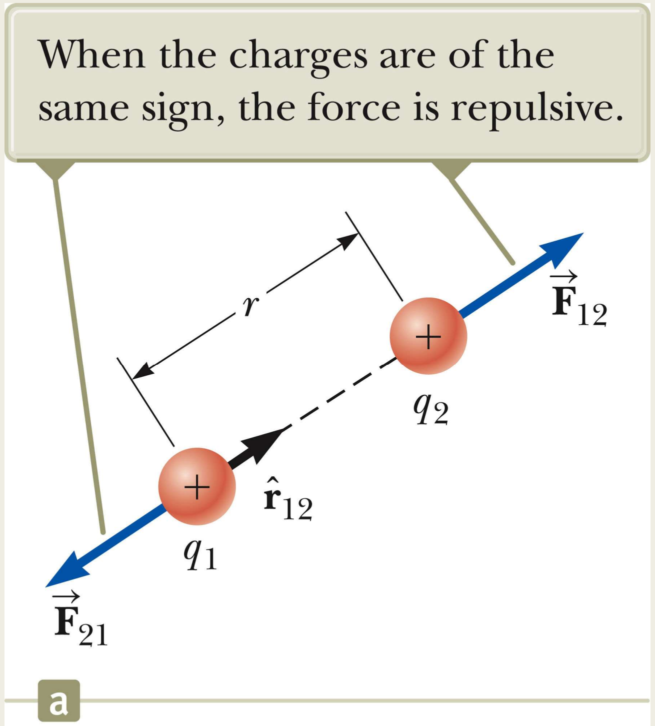
TABLE 23.1 *Charge and Mass of the Electron, Proton, and Neutron*

Particle	Charge (C)	Mass (kg)
Electron (e)	$-1.602\ 176\ 5 \times 10^{-19}$	$9.109\ 4 \times 10^{-31}$
Proton (p)	$+1.602\ 176\ 5 \times 10^{-19}$	$1.672\ 62 \times 10^{-27}$
Neutron (n)	0	$1.674\ 93 \times 10^{-27}$

- The electron and proton are identical in the magnitude of their charge, but very different in mass.
- The proton and the neutron are similar in mass, but very different in charge.

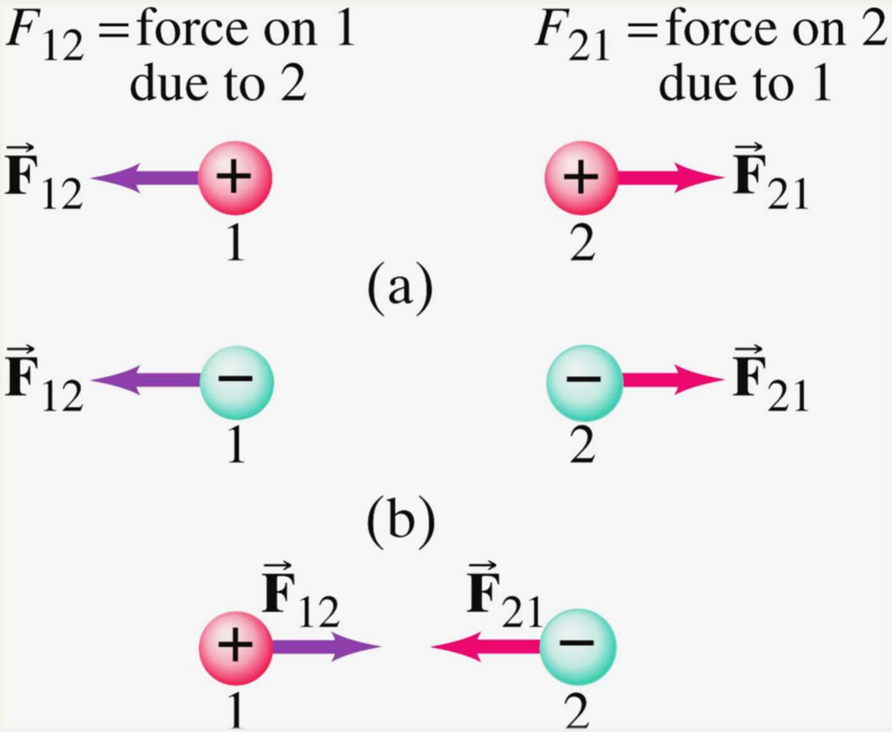
Coulomb's Law

- Coulomb's law describes the experimental results for the force of interaction between two point charges at rest, as
- The forces acting between two stationary point charges are:
- inversely proportional to the square of the separation r between the charged particles.



Coulomb's Law

- directly proportional to the product of the charges q_1 and q_2 on the two particles.
- attractive if the charges are of opposite signs and repulsive if they have the same sign.
- The forces act along the line joining the two point charges.



Coulomb's Law, Equation

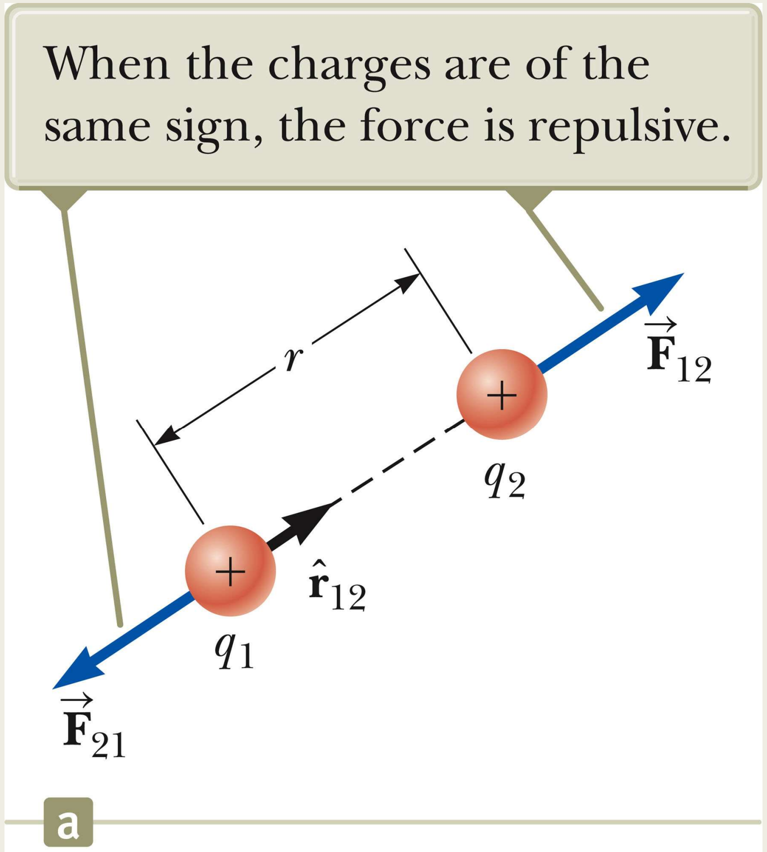
■Mathematically,

$$F_e = k_e \frac{|q_1||q_2|}{r^2} = \frac{|q_1||q_2|}{4\pi\varepsilon_o r^2}$$

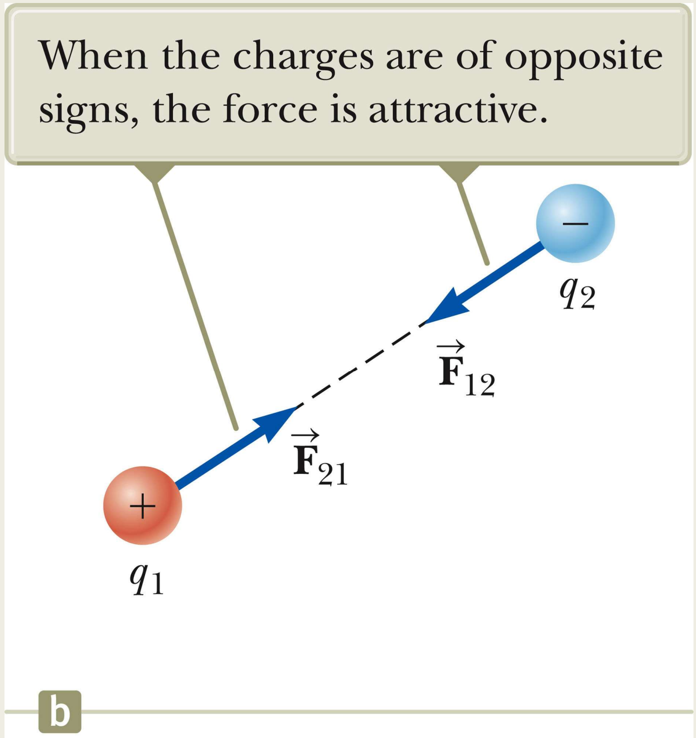
■The SI unit of charge is the **coulomb**.

■k_e is called the **Coulomb constant.**

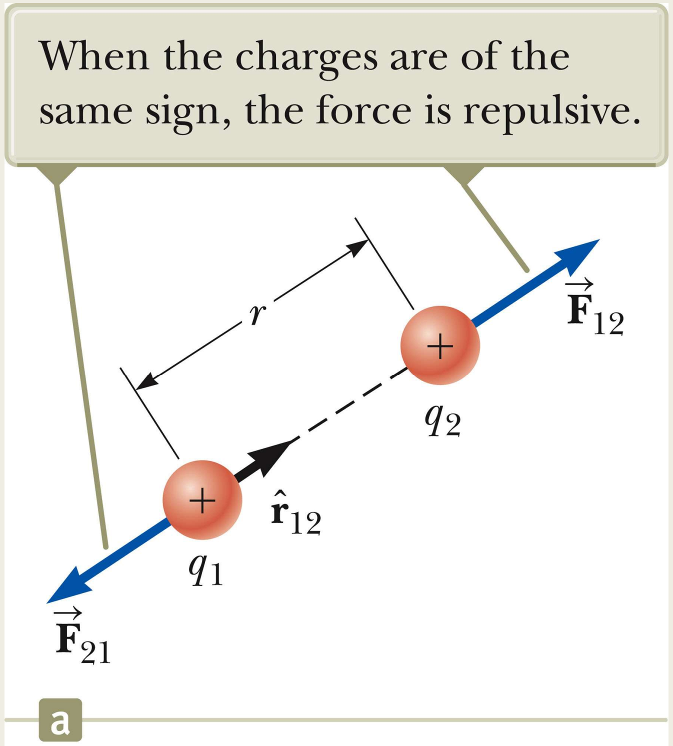
- $k_e = 8.9876 \times 10^9\ N{\cdot}m^2/C^2 = 1/(4\pi e_o)$
- e_o *is the* ***permittivity of free space.***
- $e_o = 8.8542 \times 10^{-12}\ C^2 / N{\cdot}m^2$



Vector Nature of Electrical Forces



$$\vec{F}_{21} = -\vec{F}_{12}$$



Example 1 :

- The electron and proton of a hydrogen atom are separated (on the average) by a distance of approximately $5.3x10^{-11}$ m. Find the magnitude of the electrostatic force.

$$F_e = k_e \frac{|e||-e|}{r_{12}^2} = (8.99x10^9 \, N.m^2 / C^2) \frac{(1.6x10^{-19} \, C)^2}{(5.3x10^{-11} \, m)^2}$$

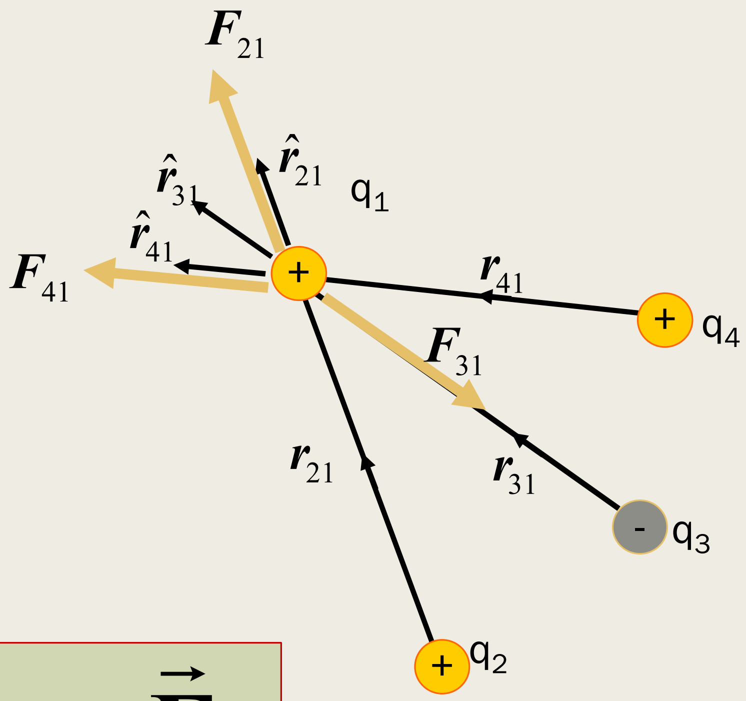
$$\therefore F_e = 8.2x10^{-8} \, N$$

The Principle of Superposition of Electrostatic Forces:

- Coulomb's law describes the forces between two point charges.
- To find the resultant force acting on one point charge due to a group of stationary point charges we need to apply the Principle of Superposition of Electrostatic Forces.
- Principle of Superposition of Electrostatic Forces states that the force on one point charge in the presence of a number of stationary point charges is the vector sum of the forces acting on the point charge due to each individual charge of the group as given by Coulomb's law.

The Principle of Superposition of Electrostatic Forces:

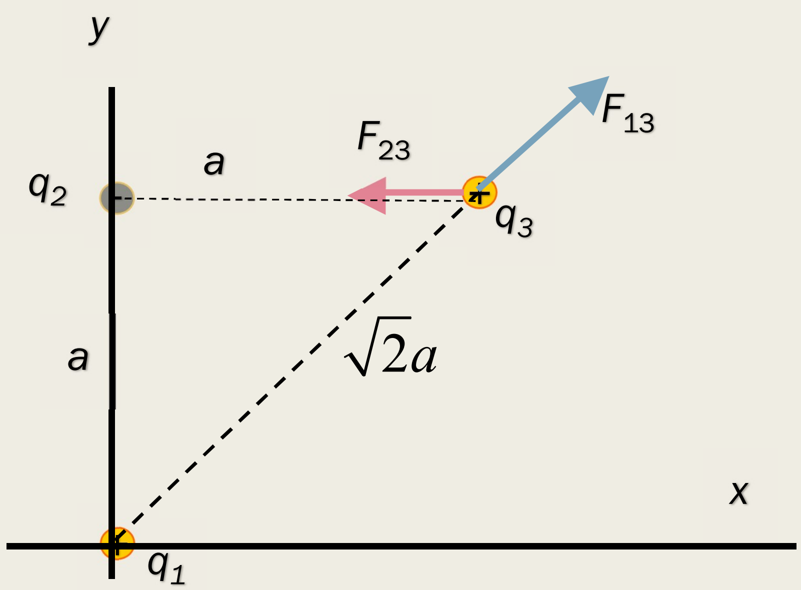
- The resultant force on q_1 is the vector sum of all the forces exerted on it by other charges.
- For example, if four charges are present, the resultant force on one of these equals the vector sum of the forces exerted on it by each of the other charges.



$$\vec{\mathbf{F}}_1 = \vec{\mathbf{F}}_{21} + \vec{\mathbf{F}}_{31} + \vec{\mathbf{F}}_{41}$$

Example 2 :

Find the resultant force acting on q_3 in the figure.



$$a = 0.10\ m,\ q_1 = q_3 = 5.0\ mC,\ q_2 = -2.0\ mC$$

Solution

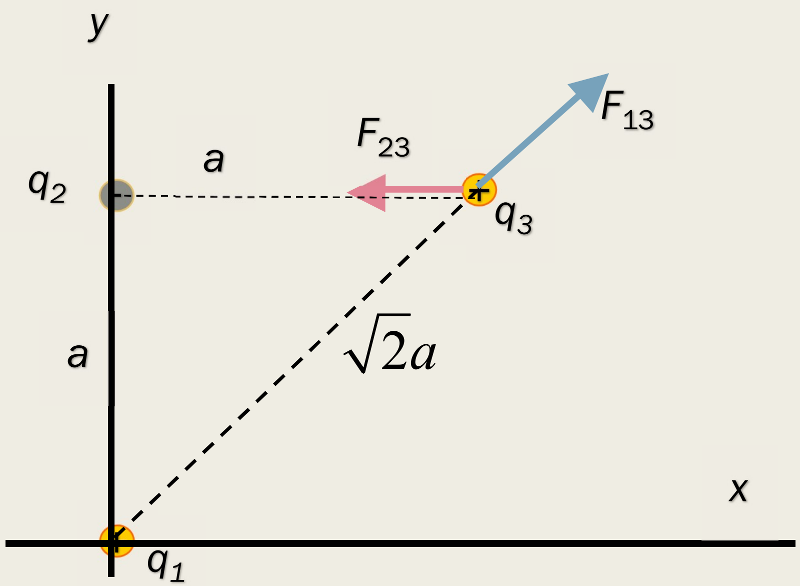
$$F_{13} = k_e \frac{|q_1||q_3|}{(\sqrt{2}a)^2} = 11\text{N}$$

$$F_{23} = k_e \frac{|q_2||q_3|}{a^2} = 9.0\text{N}$$

$$\boldsymbol{F}_3 = \boldsymbol{F}_{13} + \boldsymbol{F}_{23}$$

$$F_{3y} = F_{13y} + F_{23y} = F_{13}\sin 45^\circ + 0 = 7.9N$$

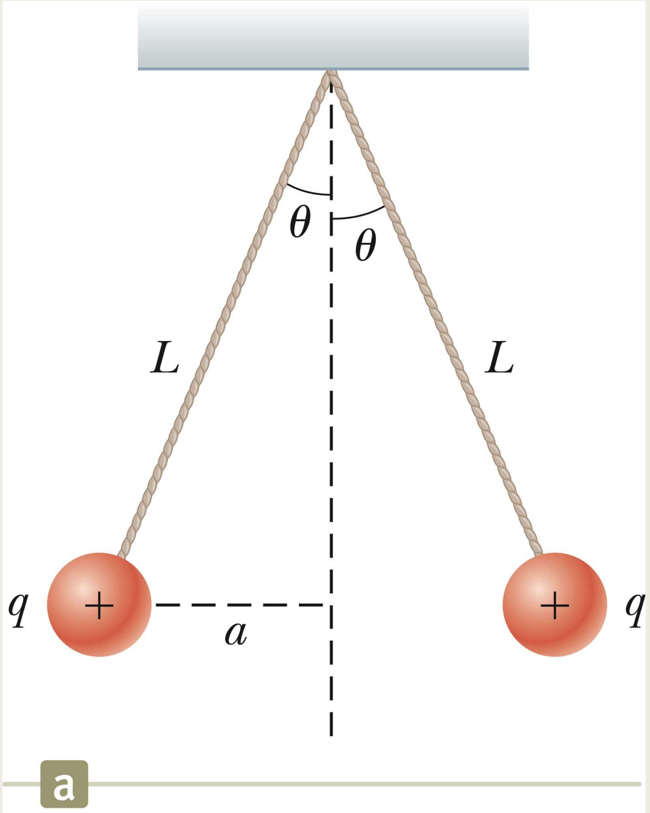
$$F_{3x} = F_{13x} + F_{23x} = F_{13}\cos 45^\circ - F_{23} = -1.1N$$



$$\therefore \boldsymbol{F}_3 = (-1.1\hat{\boldsymbol{i}} + 7.9\hat{\boldsymbol{j}})N$$

Example 3:

Two identical small charged spheres, each having a mass of 3.0×10^{-2} kg hang in equilibrium as shown in figure. The length of each string is 0.15 m, and the angle θ is 5.0°. Find the absolute value of the charge on each sphere.



Solution

$$\sum F_x = T\sin\theta - F_e = 0$$

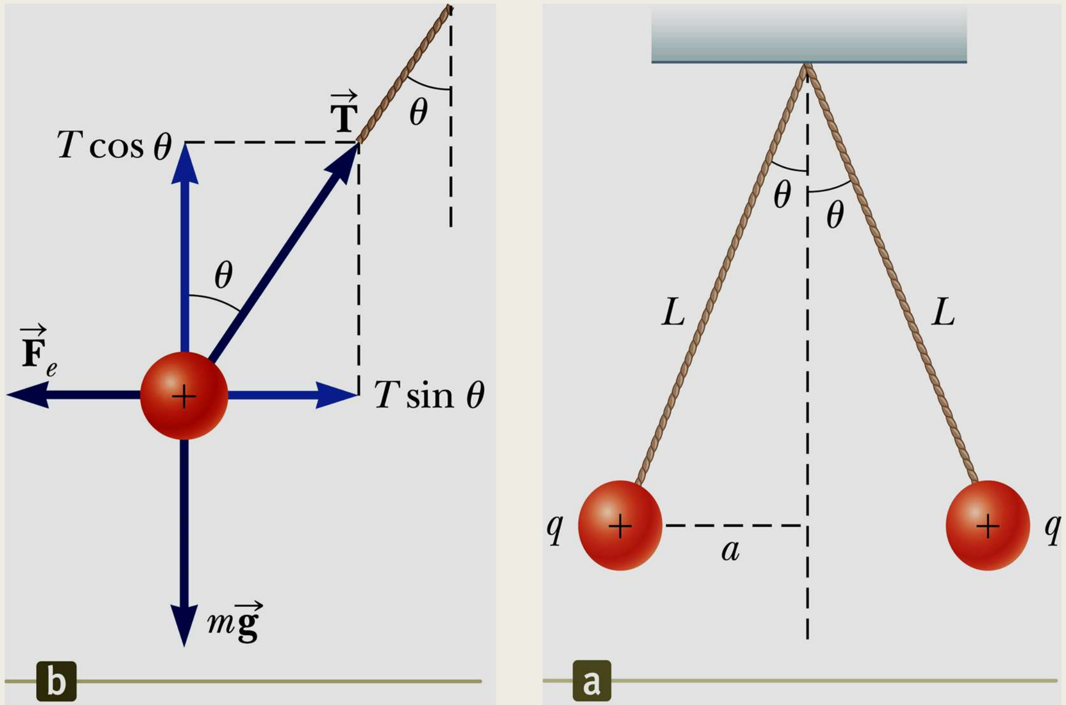
$$\sum F_y = T\cos\theta - mg = 0$$

Eliminating T from the two equations

$$\therefore F_e = mg\tan\theta = 2.6\text{x}10^{-2}N$$

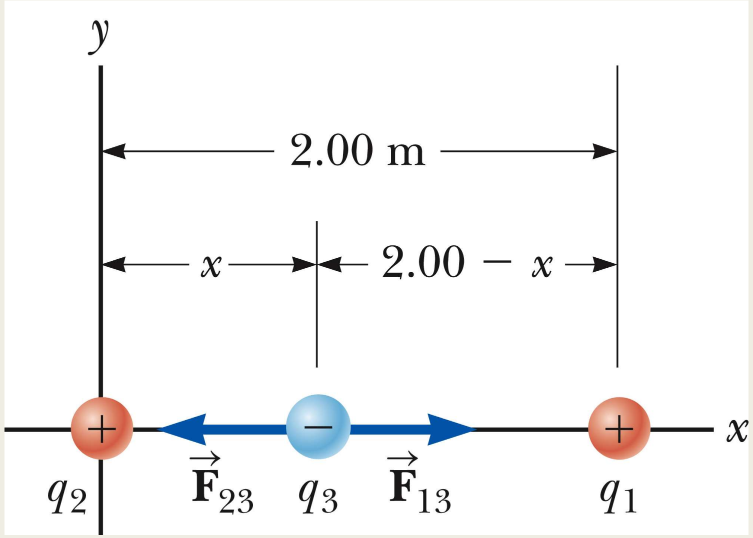
From Coulomb's law: $F_e = k_e \dfrac{q^2}{4a^2}$

$$\boxed{\therefore |q| = 4.4x10^{-8}\,\text{C}}$$



Example 4 :

Three point charges lie along the x axis as shown in figure. The positive charge q_1 = 15.0 mC is at x = 2.00 m, the positive charge q_2 = 6.00 mC is at the origin, and the resultant force acting on q_3 is zero. What is the x coordinate of q_3.



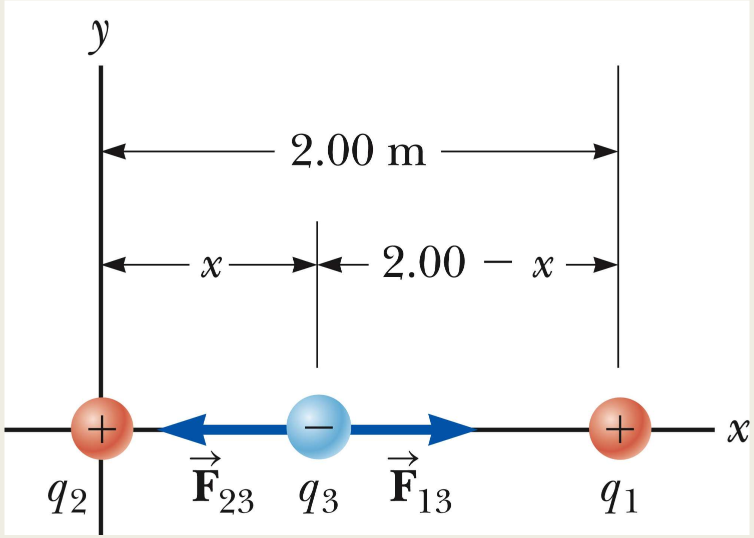
Solution

$$\therefore F_{13} = k_e \frac{|q_1||q_3|}{(2.00-x)^2} \text{ and } \qquad F_{23} = k_e \frac{|q_2||q_3|}{x^2}$$

For the net force on q_3 to be zero, then $F_{13} = F_{23}$

$$\therefore k_e \frac{|q_1||q_3|}{(2.00-x)^2} = k_e \frac{|q_2||q_3|}{x^2} \qquad \therefore q_2(2.00-x)^2 = q_1 x^2$$

This equation can be reduced to $3x^2 + 8x - 8 = 0$

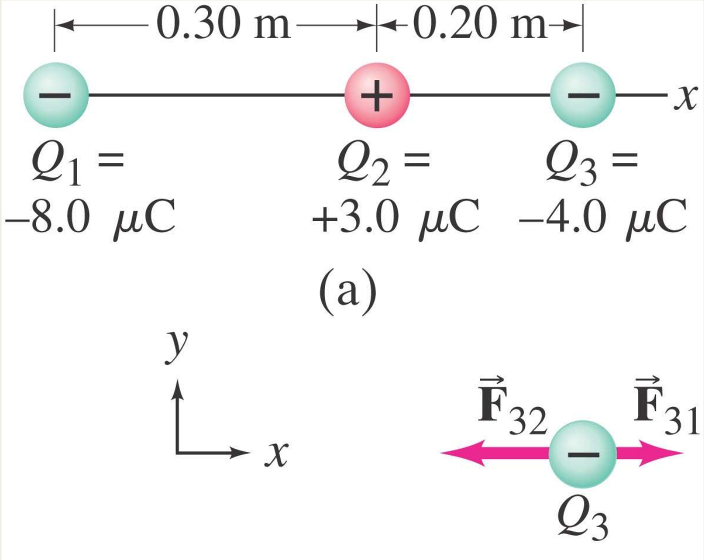


The two roots are:

- "x=0.775m" which is the location at which the net force on q_3 is zero, Accepted
- and " x= - 3.44m" which is a location at which the two forces have equal magnitudes but are in the same direction, Refused

Example 5:

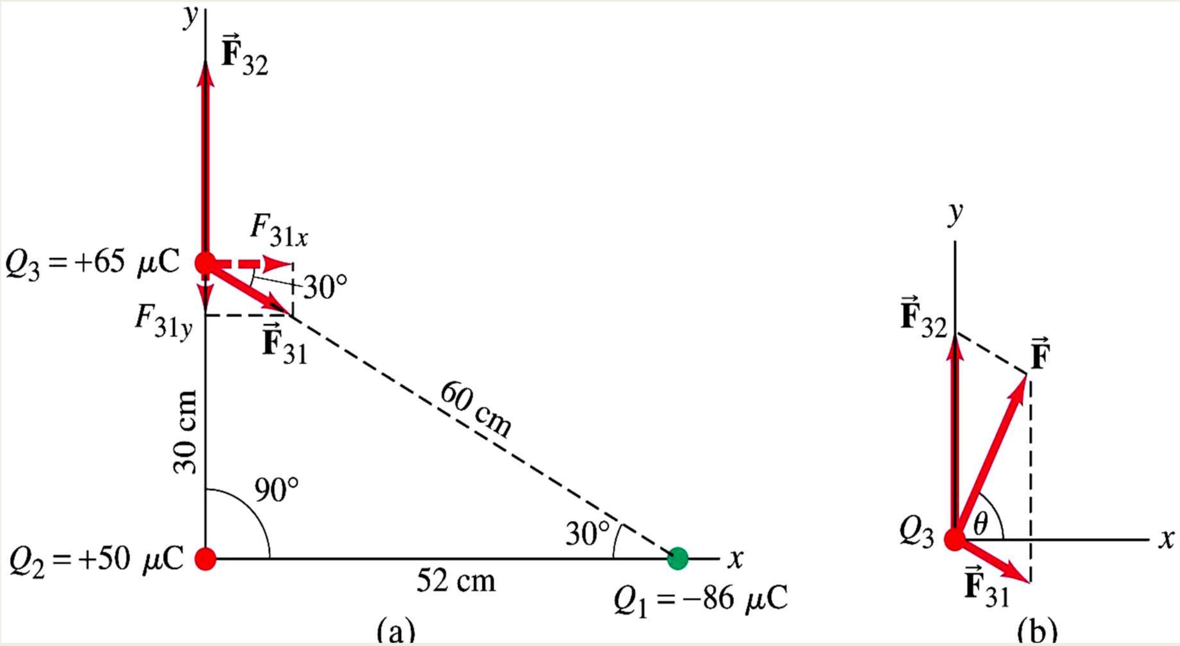
- Three charged particles are arranged in a line, as shown. Calculate the net electrostatic force on particle 3 (the -4.0 μC on the right) due to the other two charges.



F = -1.5 N (to the left).

Example 6:

- Calculate the net electrostatic force on charge Q_3 shown in the figure due to the charges Q_1 and Q_2.



F = 290 N, at an angle of 65°